

Challenge (Habanero)

Q1. Factor the expression: $x^2 + 3x + 2x + 6$

- (A) $x(x + 3) + 2(x + 3)$
- (B) $x(x + 2) + 3(x + 2)$
- (C) $x(x + 1) + 2(x + 3)$
- (D) $x(x + 4) + 2(x + 1)$
- (E) $x(x + 2) + 3(x + 1)$

Q2. A park ranger is tracking the population of rabbits in a forest. The population is given by the expression: $2x^2 + 5x + 2x + 5$. Factor this expression.

- (A) $2x(x + 2) + 5(x + 1)$
- (B) $2x(x + 1) + 5(x + 2)$
- (C) $2x(x + 3) + 5(x + 1)$
- (D) $2x(x + 5) + 2(x + 1)$
- (E) $(x + 1)(2x + 5)$

Q3. A camper is planning a trip and needs to calculate her fuel consumption. The total fuel consumption is given by: $3x^2 + 2x + 5x + 10$. Factor this expression.

- (A) $3x(x + 2) + 5(x + 2)$
- (B) $3x(x + 5) + 2(x + 1)$
- (C) $3x(x + 1) + 2(x + 5)$
- (D) $3x(x + 10) + 2(x + 1)$
- (E) $(3x + 5)(x + 2)$

Q4. Factor the expression: $x^2 + 4x + 3x + 12$

- (A) $x(x + 3) + 4(x + 3)$
- (B) $x(x + 4) + 3(x + 4)$
- (C) $x(x + 1) + 3(x + 4)$
- (D) $x(x + 7) + 3(x + 1)$
- (E) $x(x + 4) + 3(x + 3)$

Q5. A group of friends are planning a road trip. The cost of fuel for the trip is given by the expression: $2x^2 + 5x + 3x + 15$. Factor this expression.

- (A) $2x(x + 3) + 5(x + 3)$
- (B) $2x(x + 5) + 3(x + 1)$
- (C) $2x(x + 1) + 3(x + 5)$
- (D) $2x(x + 15) + 3(x + 1)$
- (E) $2x(x + 3) + 5(x + 3)$

Q6. Factor the expression: $3x^2 + 2x + 5x + 10$

- (A) $3x(x + 2) + 5(x + 2)$
- (B) $3x(x + 5) + 2(x + 1)$
- (C) $3x(x + 1) + 2(x + 5)$
- (D) $3x(x + 10) + 2(x + 1)$
- (E) $3x(x + 2) + 5(x + 2)$

Q7. A hiker is tracking her distance traveled. The total distance is given by the expression: $x^2 + 2x + 4x + 8$. Factor this expression.

- (A) $x(x + 2) + 4(x + 2)$
- (B) $x(x + 4) + 2(x + 4)$
- (C) $x(x + 1) + 2(x + 4)$
- (D) $x(x + 8) + 2(x + 1)$
- (E) $x(x + 2) + 2(x + 4)$

Q8. Factor the expression: $2x^2 + 3x + 2x + 6$

- (A) $2x(x + 2) + 3(x + 2)$
- (B) $2x(x + 3) + 2(x + 3)$
- (C) $2x(x + 1) + 3(x + 2)$
- (D) $2x(x + 6) + 3(x + 1)$
- (E) $2x(x + 3) + 2(x + 1)$

Open Ended Questions (FRQ)

Q9. Verify the factoring of the expression $x^2 + 5x + 4x + 20$ by multiplying the factors.

Q10. Factor the expression $3x^2 + 2x + 6x + 12$ and verify the result by multiplying the factors.

Chef's Tips

- Chef's Tips: Factoring by grouping requires careful attention to the terms being grouped.
- Always verify your factoring results by multiplying the factors.
- Encourage students to use different methods, such as factoring by grouping and factoring quadratics, to solve problems.